University of Essex
MSc Artificial Intelligence
Module: KRAR_PCOM7E October 2025

Collected Unit Notes

For convenience, unit notes from this module are collected here.

Unit 1: Introduction to Knowledge Representation and Reasoning

The focus of this unit was to act as an introduction to the field of knowledge representation and reasoning, with discussion of the terms "data," "information," and "knowledge" as applied to a technical context.

I also contributed my thoughts to a collaborative discussion about the principles of knowledge representation in computing systems as they relate to the use of formal logic and symbolic reasoning in artificial intelligence platforms.

Unit 2: Sets, Set Theory, Truth Tables and Logic

The objectives of this unit were to review the fundamental principles of set theory, first order logic, and truth tables.

My takeaway was a better understanding of how to define and apply set rules and of how to use truth tables to find solutions to logic-focused problems.

Unit 3: Introduction to Reasoning

The focus of this unit was to act as a deeper dive to the field of knowledge representation and reasoning, including the fundamentals of truth tables, first order logic, logical operators, and set theory principles.

As from Unit 2, my takeaway was a better understanding of how to define and apply set rules and of how to use truth tables to find solutions to logic-focused problems.

I also summarized a collaborative peer discussion about the principles of knowledge representation in computing systems as they relate to the use of formal logic and symbolic reasoning in artificial intelligence platforms.

Unit 4: Introduction to Logic Programming

The objectives of this unit were to review programming languages based on first order logic and set theory, to write and evaluate simple statements in Prolog, and to run and evaluate simple Prolog routines.

My takeaway was a foundational understanding of both topics; in particular, a grasp of Prolog programming basics.

Unit 5: Introduction to Modelling

The objectives of this unit were to discuss the meaning of knowledge modelling and engineering and to investigate the baseline principles of knowledge engineering.

My takeaway was a better understanding of both topics.

Unit 6: Introduction to Ontology Building and Online and Offline Tools

The main objective of this unit was to use the Protégé ontology development software platform to design and develop simple ontological systems.

My takeaway was an introduction to the practical mechanics of using Protégé to design an ontological class structure.

Unit 7: Knowledge Elicitation and Formalism

The objectives of this unit were to discuss the concepts of and principles that underpin knowledge acquisition as well as formal logical structures in context of knowledge representation reasoning.

My takeaway was a better understanding of this same topic.

Additionally, I prepared a case study review of Malik, Sharan, and Hijam (2015) focused on evaluating different approaches for creating an ontology for a subject matter domain; in particular, an ontology to support text mining and text-based searches across agricultural reference works.

Reference

Malik, N., Sharan, A. and Hijam, D. (2015). *Ontology development for agriculture domain*. [online] Available at: https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=7100347 [Accessed 7 Dec. 2025].

Unit 8: Modelling with Protégé

The main objective of this unit was to continue to use the Protégé ontology development software platform to design and develop ontological systems.

My takeaway was a deeper understanding of the practical mechanics of using Protégé to design ontological class structures and hierarchies.

Unit 9: Formalism Techniques and Applications

As with the previous unit, the main objective of this unit was to continue to use the Protégé ontology development software platform to design and develop ontological systems.

And as with the previous unit, my takeaway was a deeper understanding of the practical mechanics of using Protégé to design ontological class structures and hierarchies.

I also contributed my thoughts to a collaborative discussion about selection criteria for ontology development software for the development of automated and Internet-enabled software agents, asserting that OWL 2 is the most robust and full-featured platform currently available for most use cases.

Unit 10: Reasoning with Protégé

The objectives of this unit were to discuss the architectural patterns used for knowledge based-systems and the concepts of attributes, languages, complements, and the close and open worlds assumptions in description logic.

My takeaway was a better understanding of those same topics.

Unit 11: Knowledge-based Technologies and Emerging Applications

The objectives of this unit were to discuss emerging approaches to developing ontology and knowledge-based systems and to develop familiarity with the concept of multi-domain ontologies.

I also summarized my thoughts to a collaborative discussion about selection criteria for ontology development software for the development of automated and Internet-enabled software agents, in which I asserted that OWL 2 is the most robust and full-featured platform currently available for most use cases.

For portfolio development, I developed an ontological model for an AI-driven search engine for a hypothetical collection of books and other library media assets.

Unit 12: Ontology Evaluation – Case Study

The main objective of this unit was to continue to discuss emerging approaches to developing ontology and knowledge-based systems.

My takeaway was a better understanding of that same topic.

I also prepared a reflective essay with my thoughts on my learning challenges and learning outcomes from the module.